Amashanyarazi: umuyoboro wa leta, imirasire y'izuba, cyangwa byombi

Uko ifamu ikwiye kubona amashanyarazi, icyo buri buryo butwara, n'intambwe zo kubishyiraho, bigenewe nyir'ifamu n'umuyagankuba (electrician).

> ***Ubusobanuro:*** *iyi nyandiko yahinduwe na mudasobwa (machine draft). Igomba gusuzumwa n'umuntu uvuga Ikinyarwanda kavukire mbere yo kuyiha abakozi. Iyo hari itandukaniro, kurikiza iy'Icyongereza.*

Status: inama yo ku wa 2026-10-28. Ingano y'umuriro ukenewe (loads) ni ibigereranyo [E] kugeza igihe hazahitwamo icyo igikoni gikuru (galley) kizatekesha n'ibyuma bishyushya amazi (heaters) byo mu byumba by'abashyitsi; ibiciro ni uko byatangajwe ku matariki yatanzwe.

Inama mu murongo umwe: huza ifamu n'umuyoboro wa REG (grid / réseau), wongereho sisitemu y'imirasire y'izuba ya hafi 10 kWp ifite bateri (batteries) za hafi 15 kWh zigenewe ibintu bitagomba guhagarara na rimwe, ushyushye amazi ukoresheje izuba, kandi ugumane moteri itanga umuriro (générateur) nk'ingoboka gusa (emergency backup).

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Ibyo ifamu ikoresha

Buri nyubako, n'ingufu ikoresha ku munsi usanzwe n'umuriro mwinshi ikurura icyarimwe. Uburyo: ingufu (power) za buri gikoresho zikubwe n'amasaha gikora, hanyuma hagakoreshwa diversity factor ku bikoresho bitajya bikorera rimwe byose [E].

Inyubako	Ku munsi (kWh)	Ntarengwa icyarimwe (peak, kW)	Ibisobanuro
Sitidiyo, ebyiri	16	13.8	30 A kuri buri sitidiyo, kuri 230 V (D-021); hob, heater ya L 30, n'ibiteganyirijwe AC (AC provision)
Ibyumba by'abashyitsi, bitatu	12	9	buri cyumba gifite douche yacyo (D-029): heater muri buri cyumba, cyangwa amazi ashyushywa n'izuba (solar hot water)

Inyubako	Ku munsi (kWh)	Ntarengwa icyarimwe (peak, kW)	Ibisobanuro
Igikoni gikuru (galley)	25	14	guteka ku mashanyarazi; hafi 8 kWh niba gitekesha gaz
Akabari (bar)	5	2	firigo, amatara, umuziki
Icyumba cyo kwigiramo (study room)	7	2	mudasobwa zigendanwa (laptops) 12, screen (écran), fans
Ibiro (office), ubwiherero rusange, igikoni cy'abashyitsi	9	5	
Pompe: iya borehole n'iyongera umuvuduko (booster)	5	2	zikora saa sita z'amanywa, igihe izuba riva cyane
Amatara yo hanze, Wi-Fi, CCTV	5	1	ibintu bitagomba guhagarara na rimwe
Igiteranyo cy'ifamu yose	**hafi 84**	**hafi 40**	harimo diversity; hafi 65 kWh ku munsi niba igikoni gikuru gitekesha gaz

Amakuru agena icyemezo

Ingingo	Agaciro	Aho byakuwe (source)
Guhuzwa n'umuyoboro bwa mbere (new connection), ku ntera itarenze m 37 uvuye ku murongo w'amashanyarazi	56,000 RWF, phase imwe cyangwa eshatu (monophasé cyangwa triphasé), harimo na mubazi (meter)	Politiki ya REG yo guhuza abakiriya (connection policy) [V ENG-18]
Kuva kuri m 37 kugeza kuri m 800	56,000 RWF hiyongereyeho ikiguzi cy'umurongo	REG [V ENG-18, ENG-19]
Kure kurushaho, cyangwa umuriro mwinshi kurusha uwo umurongo ushobora gutwara	umurongo wa MV (medium voltage, moyenne tension) na transformateur (transformer), byishyurwa n'umukiriya ariko bikaba umutungo wa REG	REG [V ENG-19]

Ingingo	Agaciro	Aho byakuwe (source)
Igihe ntarengwa cyo gutegereza	iminsi 5 y'akazi, amezi 2 iyo bisaba kwagura umurongo wa LV (low voltage, basse tension), amezi 3 iyo bisaba kwagura umurongo wa MV	RURA [V ENG-22]
Igiciro cy'umuriro wa REG, ubucuruzi (business)	376 RWF/kWh hejuru ya 100 kWh ku kwezi, hiyongereyeho TVA (VAT) ya 18%: hafi 444	RURA, guhera ku wa 1 Ukwakira 2025 [V ENG-20, ENG-21]
Igiciro cy'umuriro wa REG, amahoteli	239 RWF/kWh munsi ya 660,000 kWh ku mwaka, hiyongereyeho TVA: hafi 282	RURA [V ENG-20]; baza REG niba ifamu yemerewe iki giciro
Ibura ry'umuriro (outages)	hafi inshuro 17 ku mwaka, amasaha 11 yose hamwe, ku rwego rw'igihugu	Banki y'Isi (World Bank) [V ENG-23]; nta mibare ya Bugesera yabonetse
Umurongo w'i Gashora	feeder ya MV ya Kanazi	Itangazo rya REG rimenyesha ibura ry'umuriro [V ENG-29]
Izuba i Gashora	4.35 kWh ku munsi kuri buri kWp ya panneaux solaires (solar panels), hafi 3.9 mu Gushyingo no mu Kuboza	Global Solar Atlas [V ENG-24]
Umusoro ku bikoresho by'imirasire y'izuba	nta TVA kuri panneaux, inverters (onduleurs), bateri za deep-cycle, ibyuma bishyushya amazi bikoresheje izuba (solar water heaters) na pompe, kugeza ku wa 30 Kamena 2028	RRA [V ENG-25]
Uruhushya (licence)	ntirukenewe kuri sisitemu iri munsi ya 50 kW ikoreshwa aho yashyizwe (on site)	RURA [V ENG-26]
Kugurisha umuriro usagutse ku muyoboro wa REG	ntibirashoboka: net metering iracyigwa gusa	[V ENG-27]
Mazutu (diesel)	hafi 2,927 RWF kuri litiro	Kamena 2026 [V ENG-28, secondary]

Uburyo butatu

	Umuyoboro wa REG wonyine	**Imirasire y'izuba yonyine (off-grid)**	**Umuyoboro wa REG n'imirasire y'izuba (hybrid)**
Icyo wubaka	guhuzwa n'umuyoboro (connection), board yo ku kibanza (site board) n'insinga zijya kuri buri nyubako	panneaux zibariwe ku kwezi kubi kurusha ayandi, bateri nini, moteri yo mu minsi y'ibicu	guhuzwa n'umuyoboro, panneaux za hafi 10 kWp, bateri za hafi 15 kWh
Ikiguzi cyo gutangira	ni cyo gito kurusha ibindi niba umurongo uri hafi	ni cyo kinini kurusha ibindi: panneaux na bateri bya 84 kWh ku munsi	kiri hagati
Ikiguzi cyo gukoresha	buri kWh ku giciro cya hafi 444 RWF	hafi ntacyo, ariko bateri zirasaza	izuba ritanga umuriro ku manywa; umuyoboro wa REG ugatanga usigaye
Iyo umuriro wa REG ubuze	byose birahagarara	nta gihinduka	bateri zikomeza gutanga umuriro kuri Wi-Fi, CCTV, firigo, pompe n'amatara amwe
Bikwiriye iyi famu iyo	umurongo uri hafi kandi amafaranga ari make	umurongo uri kure kandi igiciro batanze (quote) kiri hejuru cyane	umurongo ushobora kugerwaho: ni byo bishoboka cyane

Icyo imirasire y'izuba izigama: buri kWp umuriro wayo ukoreshwa ku ifamu itanga hafi 1,589 kWh ku mwaka, zifite agaciro ka hafi 705,000 RWF ku giciro cy'ubucuruzi, cyangwa hafi 448,000 ku giciro cy'amahoteli [E, from ENG-20, ENG-24]. Kubera ko nta net metering ihari, hitamo ingano ya panneaux ihwanye n'ibyo ifamu ikoresha ku manywa, ntirenge aho.

Inama yacu

1. **Umuyoboro wa REG mbere.** Saba guhuzwa n'umuyoboro kuri triphasé (three-phase), mu cyiciro kitari icy'ingo (non-residential); amapoto y'amashanyarazi asanzwe anyura ku muhanda (ifoto yo ku wa 2026-10-28). Baza mu nyandiko niba tariff y'amahoteli ireba ifamu.
2. **Imirasire y'izuba ku manywa.** Panneaux za hafi 10 kWp kuri inverter ya hybrid (onduleur hybride) yashyizwe kuri zero export (nta muriro woherezwa ku muyoboro wa REG): zitanga

umuriro ukenerwa na pompe, amazi ashyushye, icyumba cyo kwigiramo n'ifunguro rya saa sita mu gikoni gikuru.

3. **Bateri z'ibikenerwa by'ingenzi nijoro.** Bateri za LiFePO4 za hafi 15 kWh zigenewe Wi-Fi, CCTV, firigo na congélateurs (freezers), pompe n'amatara y'inzira (path lights), kugira ngo umuriro nubura ntihagire ubimenya.
4. **Izuba rishyushya amazi.** Ibyuma bishyushya amazi bikoresheje izuba (solar water heaters) ku byumba by'abashyitsi no ku bwiherero rusange; sitidiyo zigumana udusanduku twazo twa water heater twa L 30 dukoresha amashanyarazi (D-024).
5. **Moteri y'ingoboka gusa.** Moteri ntoya ya mazutu (diesel) yo gukoresha iyo umuriro ubuze igihe kirekire, ishyiriweho insinga ku buryo ishobora guha umuriro board yo ku kibanza (site board); ntizigere iba isoko nyamukuru y'umuriro (main supply).
6. **Ibikoresha umuriro mwinshi bikore saa sita z'amanywa.** Vomesha pompe, shyushya amazi kandi umese ku manywa, igihe panneaux zitanga umuriro w'ubuntu.

Intambwe

1. Jyana indangamuntu (ID), icyangombwa cy'ubutaka (land title) n'uruhushya rwo kubaka (construction permit) ku ishami rya EUCL ry'i Bugesera, usabe igiciro cyanditse cyo guhuzwa n'umuyoboro (connection quote): triphasé, hafi 40 kW, n'intera kuva ku murongo w'amashanyarazi uri hafi kugeza ku irembo.
2. Hitamo icyo igikoni gikuru kizatekesha (gaz cyangwa amashanyarazi): bihindura igiteranyo cya buri munsi ho hafi 17 kWh.
3. Saba abashyiraho sisitemu (installers) batatu bafite uruhushya rwa RURA (class D ni yo irimo imirasire y'izuba) kuguha igiciro (quote) cya sisitemu ya hybrid ya 10 kWp ifite bateri za LiFePO4 za 15 kWh, zero export, iri munsi ya 50 kW, idasoreshwa TVA (VAT-exempt).
4. Saba Ikigo cy'Imisoro n'Amahoro (RRA), mu nyandiko, kwemeza ko bateri za LiFePO4 zibarwa nka deep-cycle, kugira ngo zikomeze kudasoreshwa TVA.
5. Umuyagankuba ategura board yo ku kibanza: feeder imwe ya 32 A kuri buri sitidiyo, feeders z'izindi nyubako zose, inverter, na change-over ya moteri (igikoresho gihinduranya umuriro wa REG n'uwa moteri).
6. Hitamo ufite ibiciro (quotes) mu ntoki, ukurikije ibyatuma inama ihinduka biri hepfo, kandi wandike icyemezo mu gitabo cy'ibyemezo (decision log).

Ibyatuma inama ihinduka

- Igiciro cyo guhuzwa n'umuyoboro (connection quote) kirenze cyane ikiguzi cya panneaux na bateri z'inyongera zakenerwa kugira ngo ifamu ibe off-grid: hitamo imirasire y'izuba yonyine.
- Net metering itangiye gukurikizwa: ongeraho panneaux, ugurishe umuriro usagutse.
- Igikoni gikuru gitekesha gaz: sisitemu ya hybrid ishobora kugabanywa, ikagera kuri hafi 8 kWp.

- Ibura ry'umuriro ku ifamu rigaragaye ko rimara igihe kirekire kurusha imibare y'igihugu: ongera ubushobozi bwa bateri (battery capacity).